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Clinical Practice Guidelines for Brain Metastasis From Solid Tumors

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With advancements in systemic cancer therapies, the incidence and diagnosis of brain metastases (BMs) have increased, necessitating updated clinical practice guidelines. There also were developments in neurosurgical and radiotherapeutic modalities for intracranial lesions, as well as advances in immune and targeted therapies for BMs of systemic cancers. Recently, the ASCO-SNO-ASTRO and EANO-ESMO have published clinical guidelines for BMs from solid tumors. The ASCO-SNO-ASTRO guidelines, published in 2021, underwent a systematic literature review and critical evaluation by their Expert Panel, addressing the key questions in various therapies such as surgery, radiotherapy, and systemic therapy for the recommendations. Similarly, the EANO-ESMO guidelines, also published in 2021, involved a selection of relevant literature by expert authors, with final references confirmed through consensus, focusing on prevention, diagnosis, therapy, and follow-up. This review aims to provide an overview of the recent clinical practice guidelines for BMs from solid tumors, based on these two recently developed guidelines.

Keywords Brain metastases; Malignant brain neoplasm; Brain tumor.

INTRODUCTION

Brain metastases (BMs), secondary tumors originating from solid organs outside the central nervous system (CNS), post a significant and increasing challenge in neuro-oncology. BMs are more common than primary brain tumors, constituting over half of all clinically diagnosed brain tumors in adults [1]. The approach to treating BMs has significantly advanced over recent decades. Initially, in the 1970s, palliative measures such as steroids and whole brain radiation therapy (WBRT) were recommended [2]. However, with advancements in systemic cancer control, the treatment perspective for BMs has evolved, driven by targeted therapies for primary solid organ cancers and the development of neurosurgical modalities. This paper provides an overview of the clinical guidelines for BMs from solid tumors, focusing on the latest ASCO-SNO-ASTRO (American Society of Clinical Oncology–Society for Neuro-Oncology–American Society for Radiation Oncology) guideline [3] and EANO-ESMO (European Association of Neuro-Oncology–European Society for Medical Oncology) guideline [4], which excluded lymphoma, leukemia, and BMs from primary brain tumors (Table 1).

GUIDELINE DEVELOPMENT PROCESS

The development of the ASCO-SNO-ASTRO guideline for BMs from solid tumors involved a multidisciplinary Expert Panel, including an ASCO guidelines staff member with expertise in health research methodology. This panel conducted a systematic literature review, encompassing randomized trials and other nonrandomized evidences from January 2008 to April 2020. Additionally, they included selected randomized trials published before 2008, based on the expert opinions of the panel. This process included critical review and evidence interpretation to formulate guideline recommendations. The SNO guideline committee and the ASTRO Board of Directors reviewed and approved this guideline. A set of clinical questions for each therapy form was considered (Table 2), using evidence types (evidence-based and informal consensus),

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Table 1. Overview of the two clinical guidelines for brain metastases from solid tumors

	ASCO-SNO-ASTRO guideline	EANO-ESMO guideline
Relevant literatures	Systematic review (including randomized trials and other nonrandomized evidences, published from January 2008 to April 2020) Nonsystematic review (randomized trial of great importance, published prior to 2008)	PubMed searches (published from January 2011 to August 2020), with final references selected based on author consensus, emphasizing on originality and relevance
Guideline development methodology	After analyzing the literatures, guidelines are provided with supporting evidence wherever possible	In addition to supporting evidences, recommendations for some cases with low evidence levels are also based on expert opinion and consensus
Scope of guideline	Includes surgery, systemic therapy, radiation therapy, and the timing and interaction of these therapies	Encompasses diagnosis, treatment (surgical, systemic, radiation therapy, and integration of therapies), monitoring, and follow-up

quality of evidence (Evi.: high, intermediate, low, and insufficient), and strength of recommendation (Rec.: strong, moderate, and weak) (Table 3).

The EANO-ESMO guidelines were developed by the ESMO Guidelines Committee following their standard operation procedures. Expert authors selected relevant literature published from January 2011 to August 2020 through PubMed searches. The final references were confirmed by author consensus, focusing on originality and relevance. The guideline applied level of evidence (Class I, II, III, and IV) and grades of recommendation (Level A, B, and C) using criteria from the European Federation of Neurological Societies and Infectious Disease Society of America-United States Public Health Service Grading System, as recommended by EANO and ESMO (Table 4). Recommendations for some low evidence level cases were based on expert opinion and consensus, rather than evidence from clinical trials evidence.

CLINICAL PRACTICE GUIDELINES

Diagnostic procedures

The EANO-ESMO guidelines present recommendation for diagnosing BMs, incorporating neuroimaging, pathology, and liquid biopsies. Key recommendation include: mandatory neuroimaging for cancers patients with the new neurologic symptoms and signs (EANO: III, B; ESMO: III, B); for higher rate of detection of asymptomatic BMs, screening neuroimaging for subgroups of cancer patients with higher risk of BMs. The subgroups of cancers included the lung cancer with the possible exception of stage I non-small-cell lung cancer (NSCLC), stage IV melanoma, and metastatic human epidermal growth factor receptor 2 (HER2)-positive and triple-negative breast cancer (EANO: IV, n/a; ESMO: IV, B) [5,6]. Diagnostic neuroimaging should include brain magnetic resonance imaging (MRI) with pre- and post-contrast T1-weighted, T2-weighted and/or T2-FLAIR and diffusion-weighted sequences (EANO: III, C; ESMO: IV, B). The histopathological and immunohistochemical work-up of BMs should be performed (EANO: IV, n/a; ESMO: V, B). The BM specimens should be reconfirmed the treatment-relevant predictive biomarkers detected in the primary tumor or extra-CNS metastasis (EANO: IV, n/a; ESMO: V, B) (Table 5). The cell-free tumor DNA in the blood or cerebrospinal fluid (CSF) analysis is not routine for BM characterization or monitoring (EANO: IV, n/a; ESMO: IV, C). CSF studies, including cytology, are recommended when the leptomeningeal metastasis is suspected on clinical or neuroimaging findings (EANO: III, C; ESMO: IV, B).

Surgical treatment

The ASCO-SNO-ASTRO and EANO-ESMO guidelines provide similar recommendations for surgical treatment of BMs, with minor differences in the method of resection (piecemeal vs. en-bloc resection). They described that surgery should considered based on individual case assessments involving the patient and a multidisciplinary team. In the ASCO-SNO-ASTRO guidelines, the surgical resection for BMs could be helpful for patients with suspected BMs without a primary cancer diagnosis and/or with large BMs causing mass effect. However, if the remaining disease is not controllable via other treatment methods, the patients with multiple BMs and/or uncontrolled systemic disease may not be helpful from surgical resection (Informal consensus, Evi.: mixed; Rec.: moderate) [7,8]. The method of resection (piecemeal vs. en-bloc) and use of laser interstitial thermal therapy remain undetermined due to a lack of large-scale prospective trials (Informal consensus, Evi.: low, Rec.: none) [9,10].

The EANO-ESMO guidelines suggest surgery for single BMs (EANO: I, A; ESMO: II, A) and/or patients with acute symptoms of increased intracranial pressure, ambiguous neoplastic nature of a brain lesion, unknown primary tumor, more than one tumor, primary tumor rarely generating BMs, or changes in molecular profile impacting clinical decision-making (EANO:

Table 2. Clinical question for each form of therapy for brain metastases from solid tumors (ASCO-SNO-ASTRO)

Type of therapy	Clinical question
Surgery	1. What are the benefits and harms of surgery in adult patients with brain metastases?
	a. Do these benefits differ for patients with newly diagnosed disease versus recurrent disease?
	b. Are there subpopulations (e.g., number of metastases, status of extra-cranial disease) of patients who do not benefit from surgery?
	2. What are the benefits and harms of laser interstitial thermal therapy?
Systemic therapy	3. What systemic therapy (chemotherapy, immunotherapy, targeted agents) options, alone or in combination, have demonstrated clinical benefits in adults with brain metastases?
	a. Are there subpopulations of patients (i.e., clinical features, biomarker status, specific form of cancer, status of extra-cranial status, receiving steroids) who benefit more or less from those options?
	b. Is there an interaction between the benefit of systemic therapy and the use/form of radiation (e.g., stereotactic radiation therapy, whole brain radiation)?
	c. Is there an interaction between the benefit of systemic therapy and the number of metastases?
	d. Do these benefits/harms differ for patients with newly diagnosed disease versus recurrent disease?
	e. Do these benefits/harms differ for patients with resected versus unresected metastases?
	f. When can systemic therapy be used without any surgery or radiation therapy?
Radiation therapy	4. What are the benefits and harms of whole brain radiation therapy in adults with brain metastases?
	a. Are there subpopulations of patients (i.e., clinical features, biomarker status, specific form of cancer, resected/unresected) who benefit more or less from those options?
	b. Is there an interaction between the benefit of whole brain radiation therapy and the number of metastases?
	c. Do these benefits differ for patients with newly diagnosed disease versus recurrent disease?
	5. What approaches have been found to mitigate the harms of whole brain radiation therapy (e.g., radio-protectants, memantine, hippocampal avoidance)?
	6. What are the benefits and harms of stereotactic radiosurgery/radiation therapy in adults with brain metastases?
	a. Are there subpopulations of patients (i.e., clinical features, biomarker status, specific form of cancer, resected/unresected) who benefit more or less from those options?
	b. Is there an interaction between the benefit of stereotactic radiosurgery and the number of metastases?
	c. Do these benefits differ for patients with newly diagnosed disease versus recurrent disease?
	d. Do these benefits and risks differ between stereotactic radiosurgery and stereotactic radiation therapy and when is either more appropriate?
	7. What are the relative benefits and harms of stereotactic radiosurgery/radiation therapy compared to whole brain radiation therapy?
	a. Do the relative benefits and harms differ in subpopulations of patients (i.e., clinical features, biomarker status, specific form of cancer, resected/unresected)?
	b. Do these benefits differ for patients with newly diagnosed disease versus recurrent disease?
	c. Is there benefit from combining whole brain radiation therapy and stereotactic radiosurgery compared to either whole brain radiation therapy or stereotactic radiosurgery alone?
	8. What are the benefits and harms of using radiation sensitizers?
Timing and interaction of therapy	9. How does the relative timing of surgery, radiation therapy, and systemic therapy affect the benefits/harms of those therapies?
	a. Are there are other important interactions between these forms of therapy?

III, C; ESMO: IV, B). The extent of resection could be associated with local control of BMs [11,12] and if possible, the en-bloc resection may result in lower risk of recurrence and leptomeningeal metastasis than piecemeal resection (EANO: I, A; ESMO: II, A) [9]. Surgery is also advised for certain BM patients suitable for immune checkpoint inhibition, to facilitate rapid steroid tapering and optimize therapeutic efficacy (EANO: III, n/a; ESMO: IV, B) [13]. Postoperative MRI within 48 hours is recommended to assess extent of resection (EANO: IV, C; ESMO: V, C).

Systemic therapy

Advancements in molecular work-up, targeted therapy, and immunotherapy have significantly improved the role of systemic therapy in treating BMs. Molecular work-ups for BMs from lung cancer, breast cancer, and melanoma are encour-

Table 3. Quality of evidence and strength of recommendation (ASCO-SNO-ASTRO)

Term	Definition
Quality of evidence	
High	High confidence that the available evidence reflects the true magnitude and direction of the net effect (e.g., balance of benefits versus harms), and further research is very unlikely to change either the magnitude or direction of this net effect.
Intermediate	Intermediate confidence that the available evidence reflects the true magnitude and direction of the net effect. Further research is unlikely to alter the direction of the net effect; however, it might alter the magnitude of the net effect.
Low	Low confidence that the available evidence reflects the true magnitude and direction of the net effect. Further research may change the magnitude and/or direction of this net effect.
Insufficient	Evidence is insufficient to discern the true magnitude and direction of the net effect. Further research may better inform the topic. Reliance on consensus opinion of experts may be reasonable to provide guidance on the topic until better evidence is available.
Strength of recommendation	
Strong	There is high confidence that the recommendation reflects best practice. This is based on: 1. Strong evidence for a true net effect (e.g., benefis exceed harms); 2. Consistent results, with no or minor exceptions; 3. Minor or no concerns about study quality; and/or 4. The extent of panelist's agreement Other compelling considerations (discussed in the guideline's literature review and analyses) may also warrant a strong recommendation
Moderate	There is moderate confidence that the recommendation reflects best practice. This is based on: 1. Good evidence for a true net effect (e.g., benefis exceed harms); 2. Consistent results with minor and/or few exceptions; 3. Minor and/or few concerns about study quality; and/or 4. The extent of panelists' agreement Other compelling considerations (discussed in the guideline's literature review and analyses) may also warrant a moderate
Weak	There is some confidence that the recommendation offers the best current guidance for practice. This is based on: 1. Limited evidence for a true net effect (e.g., benefis exceed harms); 2. Consistent results, but with important exceptions; 3. Concerns about study quality; and/or 4. The extent of panelists' agreement

aged to facilitate targeted and immune therapies. The EANO-ESMO emphasizes considering the histological and molecular characteristics of the primary tumor and previous treatments for most BM patients to select appropriate systemic therapy (EANO: IV, n/a; ESMO: IV, B). The molecular genetic work-up of BMs, rather than primary tumor, is advised for some BM patients to select targeted therapy and immunotherapy in a tumor-specific manner (EANO: IV, C; ESMO: IV, B).

The ASCO-SNO-ASTRO guidelines advise local therapy, including surgical resection, radiation therapy (RT), and stereotactic radiosurgery (SRS), for symptomatic BMs, regardless of the systemic therapy used for the systemic disease (Evidence-based, Evi.: high, Rec.: strong) [7]. The guidelines also suggest that, in patients with certain cancer types who have asymptomatic BMs, local therapy can be deferred in favor of considering systemic therapy. Similarly, the EANO-ESMO guideline also recommended the enhanced role of systemic therapy for asymptomatic or oligosymptomatic BM patients (Table 6).

For patients with asymptomatic or oligosymptomatic BMs from NSCLC, the EANO-ESMO guideline recommends treatment strategies based on the presence of actionable oncogenic driver alterations. With alterations like EGFR, ALK, or ROS1 rearrangement, upfront targeted therapies are advised (EANO: II, B; ESMO: III, B) [14,15]. In the absence of these alterations, upfront immune checkpoint inhibition alone (PD-L1 ≥50%) or combined with systemic chemotherapy (PD-L1 <50%) is recommended (EANO: II, B; ESMO: III, B) [16,17]. Conversely, the ASCO-SNO-ASTRO guideline specifies medications for asymptomatic BM patients from NSCLC, recommending osimertinib or icotinib for EGFR-mutant NSCLC (Informal consensus, Evidence: low; Recommendation: weak) [18,19], and alectinib, brigatinib, or ceritinib for ALK-rearranged NSCLC (Informal consensus, Evidence: low; Recommendation: weak) [20-22]. Additionally, for BM patients from PD-L1 expression NSCLC who are also receiving pemetrexed and a platinum agent, pembrolizumab is suggested (Informal consensus, Ev-

Table 4. Evidence classification and grades of recommendation for diagnostic measure and therapeutic intervention (EANO-ESMO)

Term	Definition
Diagnostic measure	
Evidence classification	
Class I	A prospective study in a broad spectrum of persons with the suspected condition, using a "gold standard" for case definition, where the test is applied in a blinded evaluation, and enabling the assessment of appropriate tests of diagnostic accuracy.
Class II	A prospective study of a narrow spectrum of persons with the suspected condition, or a well-designed retrospective study of a broad spectrum of persons with an established condition (by "gold standard") compared with a broad spectrum of controls, where test is applied in a blinded evaluation, and enabling the assessment of appropriate tests of diagnostic accuracy.
Class III	Evidence provided by a retrospective study where either persons with the established condition or controls are of a narrow spectrum, and where test is applied in a blinded evaluation.
Class IV	Any design where test is not applied in blinded evaluation OR evidence provided by expert opinion alone or in descriptive case series (without controls).
Grades of recommendation	
Level A	Established as useful/predictive or not useful/predictive requires at least one convincing class I study or at least two consistent, convincing class II studies
Level B	Established as probably useful/predictive or not useful/predictive requires at least one convincing class II study or overwhelming class III evidence
Level C	Established as possibly useful/predictive or not useful/predictive requires at least two convincing class III studies
Therapeutic intervention	
Evidence classification	
Class I	An adequately powered prospective, randomised, controlled clinical trial with masked outcome assessment in a representative population or an adequately powered systematic review of prospective randomised controlled clinical trials with masked outcome assessment in representative populations. The following are required: (a) randomisation concealment (b) primary outcome(s) is/are clearly defined (c) exclusion/inclusion criteria are clearly defined (d) adequate accounting for dropouts and crossovers with numbers sufficiently low to have minimal potential for bias (e) relevant baseline characteristics are presented and substantially equivalent among treatment groups or there is appropriate statistical adjustment for differences.
Class II	Prospective matched-group cohort study in a representative population with masked outcome assessment that meets (a)–(e) above or a randomised, controlled trial in a representative population that lacks one criteria (a)–(e).
Class III	All other controlled trials (including well-defined natural history controls or patients serving as own controls) in a representative population, where outcome assessment is independent of patient treatment.
Class IV	Evidence from uncontrolled studies, case series, case reports or expert opinion.
Grades of recommendation	
Level A	Established as effective, ineffective or harmful requires at least one convincing class I study or at least two consistent, convincing class II studies.
Level B	Probably effective, ineffective or harmful requires at least one convincing class II study or overwhelming class III evidence.
Level C	Possibly effective, ineffective or harmful requires at least two convincing class III studies.

idence: low; Recommendation: weak) [23].

Both guidelines recommend the ipilimumab and nivolumab combination for patients with asymptomatic or oligosymptomatic BMs from melanoma, regardless of BRAF status (ASCO-SNO-ASTRO: Informal consensus, Evidence: low; Recommendation: weak; EANO-ESMO: EANO: II, B; ESMO: II, B). Moreover, the ASCO-SNO-ASTRO guidelines suggest dabrafenib plus trametinib for patients with BRAF-V600E mutation (Informal consensus, Evidence: low; Recommendation: weak) [24,25].

For patients with asymptomatic or oligosymptomatic BMs from HER2-positive breast cancer, the EANO-ESMO guideline recommended systemic therapy with delayed WBRT for patients with preserved general status (EANO: III, C; ESMO: III, B) [26,27]. The ASCO-SNO-ASTRO guidelines advised the combination of tucatinib, trastuzumab, and capecitabine for patients with BMs who have progressed on previous trastuzumab, pertuzumab, and/or trastuzumab emtansine-based

Table 5. Predictive biomarkers for identification of the source of brain metastases

Entity	Molecular markers/targets
Breast	HER2, ER/PR, BRCA1/2, PIK3CA, PD-L1
Non-small-cell lung	EGFR, ROS1, NTRK, ALK, RET, MET, KRAS, BRAF, PD-1/PD-L1
Squamous cell	FGFR1
Melanoma	BRAF, KIT, NF1, NRAS, PD-L1
Colorectal	KRAS, BRAF, NRAS, PD-L1, MSI
Upper gastrointestinal	HER, MET
Urothelial/transitional cell	PD-L1
Endometrium	MSI
Ovarian (serous)	ER/PR, MSI
Ovarian (mucinous)	MSI

ALK, anaplastic lymphoma kinase; EGFR, epidermal growth factor receptor; ER/PR, estrogen/progesterone receptor; FGFR1, fibroblast growth factor receptor 1; HER2, human epidermal growth factor receptor 2; MSI, microsatellite instability; NF1, neurofibromin 1; PD-1, programmed cell death protein 1; PD-L1, programmed death-ligand 1; PI3KCA, phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha

therapy (Evidence-based, Evi.: low; Rec.: weak) [28,29].

The EANO-ESMO guidelines also outline systemic therapy recommendations for progressive BMs. For HER2-negative breast cancer patients with progressive BM after local treatment, standard chemotherapy options like capecitabine, eribulin, carboplatin, and bevacizumab may be considered (EANO: III, C; ESMO: III, B) [30]. For BM patients from small cell lung cancer, platinum-based chemotherapy, with or without immune checkpoint inhibition, is recommended (EANO: II, B; ESMO: II, B) [31]. For symptomatic BM patients with multiple BRAF-mutated melanoma, or those requiring 4 mg or more of dexamethasone and eligible for further treatment, dabrafenib plus trametinib is suggested (EANO: IV, B; ESMO: IV, B) [24,32].

Radiation therapy

The evolution of RT modalities and research findings has significantly altered BMs treatment strategies regarding RT. First, the decreased role of WBRT is presented. The ASCO-SNO-ASTRO guideline recommends against offering RT to asymptomatic BM patients with poor Karnofsky Performance Status (KPS); specifically, those with KPS <70 and no systemic therapy options, or KPS ≤50 (Evidence-based, Evi.: low, Rec.: moderate) [33]. The EANO and ESMO guideline advised against postoperative WBRT following neurosurgical resection or SRS (EANO: I, A; ESMO: I, E) [34,35] and supportive care without WBRT for patients with multiple BMs, ineligible for SRS, and poor KPS (EANO: I, B; ESMO: III, B) [33]. The WBRT should be considered for treatment of multiple BMs not amenable to SRS, depending on the presence of neurological symptoms, size, number and location of BMs, and the choice and availability of CNS-active systemic therapy (EANO: III, B; ESMO: III, B). Second, the SRS is increasingly used for BM treatment. The ASCO-SNO-ASTRO guideline recommends SRS alone (as opposed to WBRT or combination of WBRT and SRS) for 1 to 4 unresected BMs, excluding small cell carcinoma with tumors generally smaller than 3 or 4 cm diameter (Evidence-based, Evi.: intermediate, Rec.: moderate) [35-37]. The SRS alone should be considered to patients with 1 to 2 resected BMs, if the surgical cavity can be safely treated and considering the extent of remaining intracranial disease. This statement was based on the randomized trials between single fraction SRS and WBRT (Evidence-based, Evi.: intermediate, Rec.: moderate) [38,39]. The SRS, WBRT, and the combination of SRS plus WBRT are all reasonable options for patients with more than 4 unresected or more than 2 resected BMs and better performance status, especially for patients with KPS ≥70. SRS may be preferred for patients with better prognosis or where systemic therapy that is known to be active in the CNS is available (Informal consensus, Evi.: intermediate, Rec.: moderate) [33,40]. The EANO-ESMO guideline advised an extended role of SRS. The SRS for patients with a limited number (1−4) of BMs (EANO: I, A; ESMO: I, A) and with a higher number of BMs (5−10) and a cumulative tumor volume <15 mL (EANO: II, B; ESMO: II, B) may be considered [41,42]. The SRS to the resection cavity is recommended after complete or incomplete resection of BMs (EANO: I, A; ESMO: I, A) [43].

The ASCO-SNO-ASTRO guideline suggests memantine and hippocampal avoidance for patients receiving WBRT without hippocampal lesions and expected survival ≥4 months (Evidence-based, Evi.: high, Rec.: strong) [40,44]. However, they do not recommend radiation sensitizing agents (Evidence-based, Evi.: low, Rec.: strong) [45,46]. The EANO-ESMO discusses prophylactic cranial irradiation, still recommended for patients with limited and extended-stage small-cell lung cancer responding to chemoradiotherapy (EANO: I, A; ESMO: I, A) [47-49].

Integrated therapeutic approach, monitoring, and follow-up

The ASCO-SNO-ASTRO guideline did not specify a particular sequence of therapy for patients receiving both RT and surgery (Informal consensus, Evi.: low, Rec.: none). The EANO-ESMO emphasize the importance of multidisciplinary approach and the need for randomized trials. They recommend individualized estimations of contributions from surgery, radiation oncology and medical oncology (EANO: IV, n/a; ESMO: V, B). The therapeutic decisions should be discussed at a dedicated

Table 6. The recommendation of systemic therapy for brain metastases: ASCO-SNO-ASTRO and EANO-ESMO guideline

Type of cancer	ASCO-SNO-ASTRO guideline recommendation	Evidence rating
For patients with asymptomatic BMs, deferral of local therapy is specially recommended. This deferral should continue until evidence of intracranial progression is observed. The decision to delay local therapy should be based on a multi-disciplinary discussion, weighing potential benefits and harms.		Evidence-based Evi.: High Rec.: Strong
NSCLC	1. Osimertinib or icotinib may be offered for BM patients from EGFR-mutant NSCLC [18,19].	Informal consensus Evi: Low Rec: Weak
	2. Alectinib, brigatinib or ceritinib may be offered to patients from ALK-rearranged NSCLC [20–22].	Informal consensus Evi: Low Rec: Weak
	3. Pembrolizumab may be offered for BM patients from immunotherapy-naïve PD-L1 expression NSCLC who are also receiving pemetrexed and a platinum agent [23].	Informal consensus Evi: Low Rec: Weak
Melanoma	Ipilimumab plus nivolumab (for all patients regardless of BRAF status) or dabrafenib plus trametinib (for patients with BRAF-V600E mutation) may be offered for BM patients from melanoma [24,25,32].	Informal consensus Evi: Low Rec: Weak
Breast cancer	The combination of tucatinib, trastuzumab, and capecitabine are recommended for HER2-positive metastatic breast cancer patients with BMs who have progressed on previous trastuzumab, pertuzumab, and/or trastuzumab emtansine-based therapy [28,29].	Evidence-based Evi: Low Rec: Weak
Type of cancer	**EANO-ESMO guideline recommendation**	**Evidence rating**
For asymptomatic or oligosymptomatic BM patients, the systemic therapy is recommended (upfront systemic therapy or delay whole brain radiation therapy).		
NSCLC	Asymptomatic or oligosymptomatic BM patients without actionable oncogenic driver alterations should receive upfront immune checkpoint inhibition alone (PD-L1 ≥50%) or combined with systemic chemotherapy (PD-L1 <50%) [16,17].	EANO: II, B ESMO: III, B
	Asymptomatic or oligosymptomatic BM patients with actionable oncogenic driver alterations (EGFR or ALK or ROS1 rearrangement) should receive upfront targeted therapy [14,15].	EANO: II, B ESMO: III, B
Melanoma	Ipilimumab and nivolumab combination is the preferred first-line treatment for asymptomatic patients, regardless of BRAF mutation status (BRAF wild-type and BRAF-mutated) [24,25].	EANO: II, B ESMO: II, B
Breast cancer	Systemic therapy is advised to delay WBRT in asymptomatic or oligosymptomatic HER2-positive patients with preserved general status [26,27].	EANO: III, C ESMO: III; B
Recommendation of systemic therapy for other types of BMs (SCLC, HER2-negative breast cancer, symptomatic melanoma)		
Breast cancer	For HER2-negative patients with progressive BM after local treatment, standard chemotherapy options like capecitabine, eribulin or carboplatin and bevacizumab may be considered [30].	EANO: III, C ESMO: III, B
SCLC	Platinum-based chemotherapy, with or without immune checkpoint inhibition, is recommended [31].	EANO: II, B ESMO: II, B
Melanoma	Patients with multiple symptomatic BRAF-mutated BMs or those requiring 4 mg dexamethasone or more eligible for further treatment are recommended to receive dabrafenib plus trametinib [24,32].	EANO: IV, B ESMO: IV, B

Local therapy, surgical and/or radiation therapy and/or radiosurgery; NSCLC, non-small cell lung cancer; BMs, brain metastases; Evi., evidence level; Rec., strength of recommendation

BM board or at a disease-specific tumor board with participation of colleagues experienced in the management of CNS tumors (EANO: IV, n/a; ESMO: V, B). The randomized trials in patients with asymptomatic or oligosymptomatic BMs should be conducted to identify the optimal combined modality treatment of systemic therapy, including TKI or immune checkpoint inhibition, with surgery or SRS (EANO: IV, n/a; ESMO: V, B).

Regular neurological examinations every 2−3 months or earlier if radiological progression is suspected or neurological symptoms or signs develop, are advised (EANO: IV, n/a; ESMO: V, B). Brain MRI should be performed at similar intervals or upon suspected neurological progression (EANO: IV, C; ESMO: IV, B). Consistent assessment of neurocognitive function and consent ability is crucial (EANO: IV, n/a; ESMO: IV, B). Ad-

vanced MRI techniques and amino acid PET, are suggested for distinguishing treatment-related changes from tumor progression (EANO: III, C; ESMO: IV, C) [50,51]. Steroids should be used only in symptomatic patients (EANO: IV, n/a; ESMO: IV, B) [50]. Primary anticonvulsant prophylaxis is not recommended (EANO: I, A; ESMO: I, A). Low molecular weight heparin is suggested for thromboprophylaxis and treatment of venous thromboembolism in BM patients (EANO: II, B; ESMO: II, C). Competency to drive should evaluated considering epilepsy, cognitive, and other neurological functions, adhering to national guidelines and laws (EANO: IV, n/a; ESMO: V, n/a).

Ethics Statement

Not applicable

Availability of Data and Material

Data sharing not applicable to this article as no datasets were generated or analyzed during the study.

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Conflicts of Interest

The author has no potential conflicts of interest to disclose.

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